

A Comprehensive Review of Demand Forecasting Techniques for Urban Food Supply Chain

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Abstract - For urban food supply chains to be resilient and sustainable, accurate demand forecasting is essential, particularly for perishable foods and dairy products that are impacted by seasonality, short shelf life, and externally driven demand variability. This review paper provides a analysis of demand forecasting methods, including statistical time-series models, machine learning techniques, deep learning architectures, and hybrid approaches, that have been published in IEEE and other reputable journals. We compare the reviewed techniques based on data characteristics, evaluation metrics, and application domains such as retail supply chains, urban food, and dairy. The review identifies significant research gaps in model explainability and real-time adaptability due to culturally driven and festival-induced demand variations. The study also propose a framework aims to bridge these gaps by integrating cultural calendars with hybrid forecasting and decision-support mechanisms.

Keywords - demand forecasting, perishable food supply chain, machine learning, time-series analysis, hybrid model.

I. INTRODUCTION

Urban food supply chains operate under increasing pressure due to rapid urbanization, rising population density, and evolving consumer consumption patterns. Ensuring the continuous availability of food in cities while minimizing shortages, waste and operational inefficiencies has become a critical challenge for supply chain planners. In this context, accurate demand forecasting serves as a foundational component of resilient and sustainable urban food supply chains, directly influencing production planning, inventory control, distribution efficiency, and waste reduction. The importance of reliable forecasting is particularly pronounced for perishable food and dairy products, where short shelf life and demand uncertainty significantly amplify the consequences of forecasting errors.

Demand forecasting in urban food systems is inherently complex due to demand volatility, seasonality, promotional effects and external disruptions. Urban consumers exhibit highly dynamic purchasing behavior influenced by festivals, weather conditions, income variability, and pricing strategies. Empirical studies have shown that demand volatility especially when combined with promotional activities can substantially degrade forecasting accuracy and propagate inefficiencies across supply chain decisions [9]. For perishable food products, even small forecasting deviations can lead to overproduction, food waste, or unmet demand, making traditional static forecasting approaches insufficient for modern urban supply chains [11]. Historically, statistical time-series models such as ARIMA, SARIMA, and exponential smoothing have been widely used for food demand forecasting due to their mathematical interpretability and low computational requirements. However, these models often struggle to capture nonlinear demand patterns and complex interactions among multiple influencing factors commonly observed in urban food markets [7]. As urban supply chains generate increasingly large and heterogeneous

datasets, these limitations have motivated the adoption of machine learning (ML) and deep learning (DL) techniques.

Recent research demonstrates that machine learning-based demand forecasting models significantly improve predictive performance by learning nonlinear relationships and integrating multiple explanatory variables across products and time horizons. ML-driven frameworks have been successfully applied to multi-objective and multi-product supply chain optimization problems, enhancing both forecasting accuracy and downstream planning decisions [5]. Deep learning architectures, including recurrent neural networks and data fusion models, further extend these capabilities by modeling long-term temporal dependencies and combining multiple data channels to improve robustness and interpretability [6], [8]. Comparative studies confirm that DL models often outperform traditional and shallow ML models in seasonal and high-variability demand scenarios common in urban food and retail supply chains [7]. In parallel, hybrid forecasting approaches which integrate classical time-series models with machine learning or deep learning techniques—have emerged as a promising direction. These models aim to balance interpretability with predictive power, making them particularly suitable for operational decision-making in food and dairy supply chains. Such hybrid strategies align with the broader paradigm of Supply Chain Management 4.0, where artificial intelligence, automation, and data-driven intelligence are central to achieving adaptive and resilient supply networks [12]. Despite these advancements, recent reviews highlight persistent challenges related to model explainability, real-time adaptability, and generalizability across diverse urban contexts [10].

Motivated by these challenges and opportunities, this review paper presents a comprehensive analysis of demand forecasting techniques for urban food supply chains, with a specific focus on perishable food and dairy products. The paper systematically reviews statistical, machine learning, deep learning, and hybrid forecasting approaches published in IEEE and other high-quality journals. The reviewed methods are compared based on data characteristics, evaluation metrics, and application domains, including urban retail, food distribution, and dairy supply chains. Furthermore, drawing insights from the comparative analysis, this study proposes a modular, data-driven demand forecasting framework to support intelligent planning and decision-making in urban food systems. By identifying existing research gaps—particularly in explainability and real-time forecasting this review aims to guide future research and practical implementations toward more resilient and sustainable urban food supply chains.

II. LITERATURE REVIEW

[1] Obaidat et al. (2025) proposes a hybrid forecasting framework for daily dairy demand that compares SARIMAX against XGBoost and classical baseline models. This paper used one-year ERP based data from Jordan. The best SARIMAX configurations achieve an average MAPE of about 5.55%, outperforming XGBoost ($\approx 7.04\%$ MAPE) and improving over ETS baselines by roughly 64% in relative error reduction, majorly for culturally seasonal products like Jameed and Shanenah. Implemented STL-based seasonal decomposition, handled lunar-calendar religious events (Ramadan, Eid), framework captures structural demand shifts that generic food or agricultural

forecasting studies often ignore. As per economic analysis, SARIMAX-based systems can reduce inventory cost by approx. 15-20%, cut product waste by 25%, demonstrating significant operational value for perishable supply chains. A key research gap highlighted is scarcity of dairy specific, culturally aware forecasting studies and hybrid statistical-ML models. Future scope for multi-site, multi-year data validation and integration of DL models.

[2] Vasanthraj et al. (2023) analyzed 92 studies (2011-2021) on Industry 4.0 technologies in food supply chains (FSCs), identifying Blockchain (53 studies), IoT (32), and RFID (30) as dominant for enhancing visibility (traceability, transparency, provenance, monitoring) and operational efficiency (cost, time, waste reduction). While the study notes ML's potential for demand forecasting in perishable FSCs via data modeling, accurate forecasting emerges as critical for waste minimization in perishable goods. Study remains limited for conceptual analysis, lack of quantitative solution models and validation using real datasets.

[3] S. K. Panda and S. N. Mohanty (2023) proposed food demand forecasting using a variety of ML and DL models on the Genpact food demand dataset. The study compared seven regressors and demonstrated that deep learning models, particularly LSTM, achieved the highest accuracy with RMSE of 18.83 and MAPE of 6.56%. Concluded that feature engineering led to more accurate and real-time demand forecasting as they implemented featured variables as lag variables and EWMA variables. Also statistical tests confirmed LSTM and Bi-LSTM are best over traditional and boosting-based models. However, the study relies majorly on historical demand features, indicating scope for incorporating external factors such as weather or socio-economic variables.

[4] N. Ul Haq Qureshi et al. (2024) worked on a multivariate deep learning approach for retail demand forecasting by using Rossman store data with over one million records. Unlike major time series analysis studies works on univariate sales series, authors integrated weather, promotional and store level features. An LSTM model optimized via grid search achieved a test MAE of 458.91 and RMSE of 635.22, whereas the proposed GRU with grid search reduced the error to a test MAE of 369.79 and RMSE of 573.54. The results demonstrate improved forecasting accuracy with GRU, and further gains were observed when training separate GRU models on weather-specific subsets (normal, rain, and snow conditions). However, the model is limited by the lack of explanatory variables identifying reasons for store closures in the dataset.

[5] Abdi et al. (2025) present a machine learning-based multi-objective optimization framework that links demand forecasting with inventory and cost optimization. Models such as Random Forest and Gradient Boosting are used to forecast demand across multiple products and periods. The results show improvements in supply chain efficiency and reduced operational costs. The forecasting model operates at an aggregate level and does not address product-specific seasonal or festive demand variations.

[6] Jahin et al. (2025) proposes an Explainable multi-channel data fusion network (MCDFN) for supply chain forecasting. This model combines CNN, LSTM and GRU to capture special and temporal patterns in time series sales data. MCDFN consistently demonstrate lower error values of MSE, RMSE, MAE, and MAPE. The reported test MSE of 23.5738 and RMSE of 4.8553 indicate improved accuracy compared to standalone models such as BiLSTM, GRU, and CNN, highlighting the advantage of multi-channel feature fusion. The model integrates explainable AI techniques like SHAP. Time and permutation features increase the model's transparency and practitioner trust. However, there are limitations: the model evaluated on a single dataset may limit its generalizability across different industries or demand patterns. Additionally, the higher model complexity and parameters count could increase computational cost in large scale applications. Despite limitations, the paper provides strong evidence that MCDFN offers superior forecasting accuracy compared to

conventional deep learning approaches and represents a meaningful advancement in explainable supply chain analytics.

[7] Ensafi et al. (2022) conduct comparative analysis of machine learning and deep learning approaches for food demand forecasting in supply chain management. The dataset is about the Genpact food demand forecasting dataset. In this paper the author evaluates five machine learning models and two deep learning models (LSTM, Bi-LSTM) to predict weak demands. The major part of study is about detailed feature engineering processes. Experimental results demonstrate that deep learning models, particularly LSTM, perform traditional regression and boosting techniques, achieving lower RMSE, RMSLE, MAE, and MAPE values. The paper also strengthens its findings through extensive statistical validation using Diebold-Mariano tests, ANOVA, and residual diagnostic. However, the study does not support external factors such as holiday, promotions or special events, which limits its ability to explain sudden demand spikes. Overall the work provides a solid benchmark for food demand forecasting and highlights the effectiveness of LSTM-based models for time-series prediction in supply chain applications.

[8] Katikar et al. (2025) Talks about how Ensemble learning and Neural network models have overall accuracy in forecasting with optimized solutions in the food supply chain by using Genpact food demand dataset. Using various evaluation matrices like REMS, MAE and MAPE for performance of various models for Ensemble Learning Models (Random Forest Regressor, Gradient Boosting Regressor, XGBoost, CatBoost) and Neural Network Models (LSTM, Bi-LSTM). Among the models tested, LSTM demonstrate the highest accuracy in forecasting, out of other machine learning algorithms. However the absence of key factors like dates, holidays, and special events in the dataset limited the models's ability to capture seasonal trends and demand spikes.

[9] Abolghasemi et al. (2020) investigates the role of volatility demand in supply chain forecasting. Using a real-world dataset consisting of 843 demands time series from the FMCG food supply chain. The volatility is measured through coefficient of variation, which affect forecasting accuracy and inventory performance. The important part of study is the decomposition of demand into baseline and promotional components followed by proposal of hybrid forecasting model that combines ARIMA for baseline demand and for promotional demand it uses piecewise regression. The paper strengthens its contribution by linking the forecasting accuracy with inventory performance for to show that improved forecasts reduce inventory and order variability.

[10] Mediavilla et al. (2022) The paper provides analysis of AI methods published in last five years (2017 to 2021) for demand forecasting. Ada-boost, ANN, ARIMA, ARIMAX, BLSTM, GB, LR, LSTM, MLP, SVM, RNN, RF, RNN, XGBoost for these models evaluate performance matrices MAPE, MASE, MAE, MSE, RAME, RMSE, and TE. The study analyze clear trends towards deep learning models mainly LSTM, MLP, and ANN due to their effectiveness in handling multivariate and noisy data. It also highlights the growing relevance of hybrid methods that combines statistical and AI based techniques. It also gives insights on research gaps in collaborative forecasting and the limited applicability of current AI models to large scale, high-dimensional data.

[11] Birkmaier et al. (2024) builds paper on prior research identifying inaccurate demand forecasting as a major driver of food waste in grocery supply chains. Existing studies highlight the role of seasonality, promotions, and limited shelf life but lack alignment between forecasting outputs and planning needs. While big data analytics and forecasting support systems are increasingly discussed, user expectations and operational integration remain under-explored. This study extends the literature by empirically examining planning expectations and demand-sensing requirements necessary for deploying data-driven forecasting systems aimed at food waste prevention.

[12] Khan et al. (2022) The literature reviewed in this paper indicates that while Industry 4.0 has significantly advanced manufacturing systems, supply

chain management has lagged in digital maturity. Research shows uneven adoption of AI-based planning, digital procurement, and smart warehousing across industries. Demand forecasting and inventory management increasingly rely on AI and deep learning, yet many firms still depend on manual or legacy systems. The paper highlights gaps in integration, workforce readiness, and real-time decision-making as major challenges for achieving fully digital Supply Chain Management 4.0

Based on the systematic analysis of the twelve reviewed studies, it is observed that in supply chain management demand forecasting is crucial need particularly in food and perishable supply chains. While several studies demonstrate strong forecasting performance using machine learning and deep learning models, research explicitly focused on perishable and dairy specific demand remains limited, with only a few works incorporating cultural, seasonal, or promotional factors. Empirical evidence consistently shows that hybrid approaches combining statistical time-series models with machine learning or deep learning techniques outperform standalone models, especially under volatile and seasonal demand conditions. Forecasting performance is strongly influenced by data characteristics, where multivariate models outperform univariate approaches when external variables such as weather, promotions, or calendar effects are available, whereas simpler models remain suitable for short term, stable demand scenarios. Across the literature, RMSE and MAPE emerge as the dominant evaluation metrics, enabling comparability across studies.

The comparative summary presented in Table 1 that shows most studies on food and retail demand forecasting rely on multivariate data and mainly use hybrid or deep learning models, typically focusing on short- to medium-term forecasting horizons. Several works report better accuracy when external factors such as weather or promotions are included, indicating that demand forecasting benefits from additional contextual information. However, very few studies explicitly model culturally driven seasonality or festival related demand, especially in perishable and dairy supply chains. In addition, many approaches prioritize forecasting accuracy without adequately addressing explainability or practical integration with supply chain operations. These observations highlight the need for forecasting frameworks that are culturally informed, interpretable, and better aligned with real-world decision-making requirements.

Ref	Area	SCM Role	Data Dimension	Forecasting Method	Time Horizon	Metrics
[1]	Dairy Supply Chain	Manufacturer	Multivariate	SARIMAX + XGBoost	Short-term	RMSE, MAPE
[3]	Food Demand Forecasting	Retailer	Multivariate	RF, XGBoost, LSTM, BiLSTM	Medium-term	RMSE, MAE, MAPE
[4]	Retail Supply Chain	Retailer	Multivariate	Weather-enhanced GRU	Short-term	RMSE, MAE
[6]	Food Demand Forecasting	Retailer	Multivariate	CNN + LSTM + GRU (MCDFN)	Short-term	RMSE, MAPE
[7]	Seasonal Item Sales	Retailer	Univariate	Stacked LSTM	Medium-term	RMSE, MAPE
[8]	Food Demand Forecasting	Retailer	Multivariate	Regression Models, ANN	Short-term	RMSE, MAE
[5]	Retail Demand	Retailer	Multivariate	Promotion-aware ML Models	Short-term	RMSE

Table 1: Comparison of Techniques and models implemented

III. Gap Analysis

The gap analysis shows that existing demand forecasting studies mainly focus on improving predictive accuracy using statistical, machine learning, deep learning, and hybrid models. As gap analysis is summarized in Table . In particular, our observations reveal that most existing approaches fail to explicitly incorporate cultural and festival calendars, which play a critical role in shaping demand patterns in region-specific and culturally driven markets.

Author & Year	Focus / Methodology	Key Contributions	Identified Research Gaps
[1] Obaidat <i>et al.</i> (2025)	Hybrid SARIMAX–XGBoost for daily dairy demand forecasting	Effectively captures seasonality and improves short-term forecasting accuracy	Limited to a single dairy case study; lacks deep learning scalability and advanced explainability
[2] Vasanthraj <i>et al.</i> (2023)	Industry 4.0 adoption in food supply chains (content analysis)	Highlights visibility, traceability, and operational efficiency	Conceptual study only; no quantitative forecasting models or empirical validation
[3] Panda & Mohanty (2023)	ML and DL regressors (RF, XGBoost, LSTM, BiLSTM)	Comprehensive benchmarking and strong feature engineering	External factors such as festivals, promotions, and weather not explicitly modeled; no interpretability
[4] Qureshi <i>et al.</i> (2024)	Weather-enhanced GRU for retail demand forecasting (Rossman)	Demonstrates accuracy gains using multivariate weather data	Limited to weather variables and single retailer; cultural and festival effects ignored
[5] Abdi <i>et al.</i>	ML-based demand forecasting integrated	Links forecasting with operational	Forecasting accuracy is secondary; lacks

Author & Year	Focus / Methodology	Key Contributions	Identified Research Gaps
(2025)	with multi-objective SC optimization	decision-making	explainable deep learning focus
[6] Jahin <i>et al.</i> (2025)	MCDFN (CNN + LSTM + GRU) with explainable AI	Achieves lowest forecasting errors with interpretability	High computational complexity; not explicitly festival-aware or region-specific
[7] Ensafi <i>et al.</i> (2022)	Seasonal item sales forecasting using ML and DL	Shows stacked LSTM outperforms traditional methods	Univariate retail data only; no supply chain or external factor integration
[8] Katikar <i>et al.</i> (2024)	Regression and DL comparison for food demand	Useful baseline comparison across models	Limited robustness analysis; lacks explainability and real-world deployment
[9] Abolghasemi <i>et al.</i> (2020)	Demand volatility and promotion-aware hybrid forecasting	Robust handling of volatile and promotional demand	Does not exploit deep learning or real-time multivariate data
[10] Mediavilla <i>et al.</i> (2022)	Review of AI methods for demand forecasting	Provides taxonomy of ML and DL techniques	No empirical validation or implementation guidance
[11] Birkmaier <i>et al.</i> (2024)	Data-driven supply chain planning for perishable food using Delphi study	Identifies demand-sensing factors and planning expectations for food waste reduction	Does not implement or evaluate forecasting models; lacks quantitative performance comparison
[12] Khan <i>et al.</i> (2022)	Supply Chain Management 4.0 state-of-the-art review	Maps SCM elements and Industry 4.0 challenges across planning, logistics, and warehousing	Conceptual roadmap only; absence of operational demand forecasting models and explainable AI integration

Table 2. Gap Analysis

IV. Proposed Framework

4.1 Motivation for the Proposed Framework

Based on the comparative analysis of existing studies (Section III), no single forecasting model effectively suits all demand characteristics. As shown in Fig. 1, different demand problems such as seasonality, volatility, non-linearity, regional variation, and explainability require different modeling approaches. Statistical models offer interpretability and stable seasonal modeling, while machine learning and deep learning models capture complex and volatile demand patterns. This motivates a hybrid, calendar-aware forecasting framework that combines these models to support culturally driven demand scenarios.

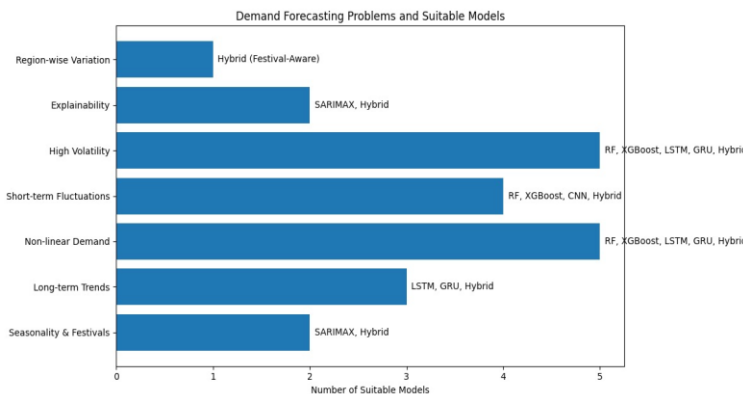


Fig. 1 Demand Forecasting Problems and Suitable Models

4.2 Hybrid Forecasting Strategy

The proposed framework adopts a hybrid forecasting strategy that combines statistical time-series models with machine learning and deep learning models to leverage their complementary strengths.

- **Statistical models (ARIMA / SARIMA / SARIMAX)** are employed to:
 - Capture long-term trend components
 - Model regular seasonality
 - Incorporate calendar-driven cyclic patterns
 - Provide interpretability and baseline forecast stability
- **Machine learning and deep learning models (RF, XGBoost, LSTM, GRU)** are used to:
 - Learn non-linear demand behavior
 - Capture sudden demand spikes and drops
 - Handle high volatility and short-term fluctuations
- **Hybrid Design Logic**

Instead of competing with each other, the models are assigned complementary roles:

- Statistical models focus on structured and predictable temporal patterns

- ML/DL models learn residual components and complex non-linear variations

➤ **Final demand forecasts are generated using:**

- Residual learning (ML/DL models trained on statistical model errors)
- Weighted ensemble fusion of multiple model outputs
- Context-based model weighting, where:
 - Higher weight is assigned to ML/DL models during festivals and volatile periods
 - Higher weight is assigned to statistical models during stable demand periods

This hybrid design directly aligns with the observations summarized in Fig. 1, where hybrid models consistently demonstrate suitability across multiple demand challenges.

4.3 Cultural and Festival-Aware Demand Modeling

➤ **Critical Limitation in Existing Approaches**

Most existing demand forecasting approaches treat festivals as implicit patterns embedded within historical sales data. This assumption is insufficient for culturally sensitive and region-specific markets, particularly in countries such as India.

For perishable food products (e.g., milk, curd, sweets), demand during festivals such as Diwali, Navratri and regional fairs is characterized by:

- Sudden and sharp magnitude shifts
- Region-dependent consumption behavior
- Non-repetitive calendar alignments due to lunar calendars

Learning demand patterns solely from historical dates without semantic festival context limits the model's ability to generalize to unseen or shifting festival scenarios.

➤ **Proposed Cultural Calendar Integration**

To address this limitation, the proposed framework explicitly incorporates a **Cultural and Festival Calendar Layer**, which includes:

- National festivals
- Regional festivals
- Religious events
- Local cultural occasions

Each calendar event is mapped to:

- Event date or date range
- Associated geographical region
- Festival intensity level (low / medium / high)
- Product relevance score

This cultural calendar data is aligned with historical production and sales records, enabling the forecasting models to distinguish between:

- Regular seasonal demand
- Festival-induced demand surges

➤ **Significance of Festival Awareness**

- Enhances demand pattern interpretability
- Improves generalization to unseen festival dates
- Reduces overfitting to date-only correlations
- Is particularly critical for perishable goods, where inaccurate forecasting directly increases food wastage

This component directly addresses a major research gap identified in Section III, namely the lack of explicit festival-aware demand modeling in existing literature.

4.4 Proposed Framework Architecture

The proposed framework consists of five interconnected layers, as illustrated in Fig. 2, designed to support accurate, robust, and interpretable demand forecasting in supply chain systems.

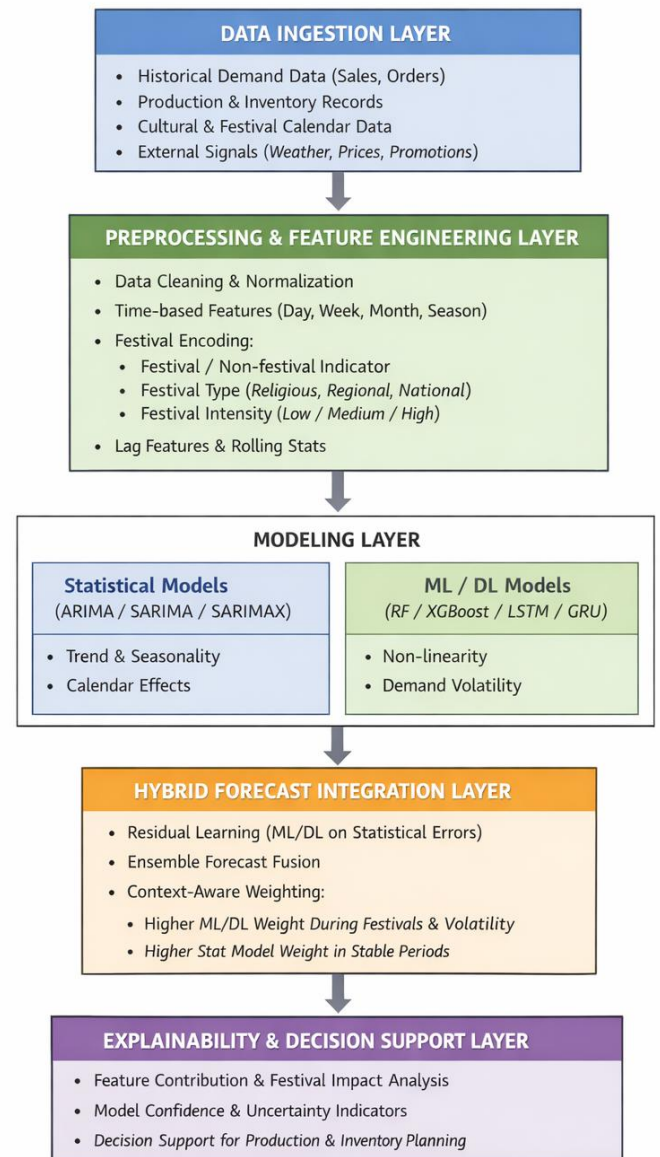


Fig. 2 Architecture diagram

The process begins with the Data Ingestion Layer, which integrates historical demand, production and inventory records, cultural and festival calendars, and optional external signals such as weather and pricing.

These inputs are transformed in the Preprocessing and Feature Engineering Layer, where time-based, lagged, and festival-aware features are extracted to capture both regular seasonality and event-driven demand patterns.

The Modeling Layer employs a dual-structure approach, combining statistical time-series models to learn trend and seasonal components with machine learning and deep learning models to capture non-linear and volatile demand behavior. Forecast outputs from these models are fused in the Hybrid Forecast Integration Layer using residual learning, ensemble fusion, and context-aware weighting that adapts to stable and festival-driven demand conditions. Finally, the Explainability and Decision Support Layer provides transparent insights into feature contributions, model confidence, and actionable recommendations to support production and inventory planning. This layered architecture ensures scalability, interpretability, and adaptability while explicitly addressing festival-driven and region-specific demand variations.

4.5 Expected Contributions of the Proposed Framework

The proposed framework contributes to both academic research and industry practice by:

1. Providing a systematic and well-justified hybrid forecasting strategy
2. Introducing explicit cultural and festival awareness into demand forecasting
3. Improving forecast reliability for perishable food supply chains
4. Reducing overproduction and food wastage through proactive planning
5. Enhancing model interpretability and operational trust

V. Conclusion

This paper presented a comprehensive review of demand forecasting methodologies applied to urban food supply chains, with particular focus on perishable food and dairy products. Study compare statistical time-series models, machine learning techniques, deep learning architectures, and hybrid approaches reported in IEEE and related high-impact journals. They were systematically analyzed and compared based on data characteristics, forecasting horizons, evaluation metrics and application domains. The analysis indicates that advanced ML and DL models improve predictive accuracy under non-linear and volatile demand conditions, but they often lack interpretability and contextual awareness required for practical supply chain decision-making. The review further demonstrates that hybrid forecasting approaches, which integrate statistical and AI-based models, offer improved robustness by combining structured seasonal modeling with non-linear learning capabilities. However, a critical research gap remains in the explicit modeling of culturally driven and festival-induced demand variations, particularly in region specific markets. Existing studies largely rely on historical demand patterns without incorporating semantic cultural context, limiting their adaptability to dynamic festival calendars and real-time demand shifts.

To address these limitations, this paper proposed a modular, data driven, and festival aware hybrid forecasting framework that integrates cultural calendar information with statistical and machine learning models, ensemble fusion strategies and explainability mechanisms. The proposed framework aims to enhance forecast reliability, support intelligent production and inventory planning and reduce food wastage in perishable supply chains.

Overall, this review provides a structured synthesis of current research, identifies key methodological gaps and outlines a scalable framework to guide future development of interpretable and context-aware demand forecasting systems for resilient urban food supply chains.

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